

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)

Python

Q. 1. Write algorithm for the following:

a. Write a program that takes two lists and returns True if they have at least one common member.

Code:

```
prac7.py - C:/Users/mvluc/Desktop/prac7.py (3.14.6)
File Edit Format Run Options Window Help

list1 = input("Enter first list : ").split()
list2 = input("Enter second list : ").split()

common = False

for i in list1:
    if i in list2:
        common = True
        break

print(common)
print("laxmi jha")
```

Output::

```
===== RESTART: C:/Users/n
Enter first list : 12345
Enter second list : 34567
False
laxmi jha
>>> |
```

```
===== RESTART: C:/Users/mvluc/Desktop/prac7.py =====
Enter first list elements (space separated): 1 2 3
Enter second list elements (space separated): 1 5 6
True
>>>
```

b. Write a Python script to sort (ascending and descending) a dictionary by value.

Code:

```
prac7.py - C:/Users/mvluc/Desktop/prac7.py (3.14.6)
File Edit Format Run Options Window Help
n = int(input("Enter number of items: "))

d = {}

for i in range(n):
    key = input("Enter key: ")
    value = int(input("Enter value: "))
    d[key] = value

asc = dict(sorted(d.items(), key=lambda x: x[1]))
desc = dict(sorted(d.items(), key=lambda x: x[1], reverse=True))

print("Ascending:", asc)
print("Descending:", desc)
```

Output:

```
>>> ===== RESTART: C:/Users/mvluc/Desktop/prac7.py =====
Enter number of items: 4
Enter key: laxmi
Enter value: 12
Enter key: shalu
Enter value: 13
Enter key: shah
Enter value: 10
Enter key: lauhh
Enter value: 23
Ascending: {'shah': 10, 'laxmi': 12, 'shalu': 13, 'lauhh': 23}
Descending: {'lauhh': 23, 'shalu': 13, 'laxmi': 12, 'shah': 10}
```

Q. 2. Write a python program that creates a list of 3 fruit names, adds one more fruit to the list, print entire list and print the first and last fruit in the list.

Code:

```
prac7.py - C:/Users/mvluc/Desktop/prac7.py (3.14.6)
File Edit Format Run Options Window Help
fruits = []

for i in range(3):
    fruit = input("Enter fruit name: ")
    fruits.append(fruit)

new_fruit = input("Enter one more fruit: ")
fruits.append(new_fruit)

print("Fruit List:", fruits)
print("First Fruit:", fruits[0])
print("Last Fruit:", fruits[-1])
```

Output:

```
RESTART: C:/Users/mv1uc/Desktop/p1a
Enter fruit name: apple
Enter fruit name: banana
Enter fruit name: kiwi
Enter one more fruit: mango
Fruit List: ['apple', 'banana', 'kiwi', 'mango']
First Fruit: apple
Last Fruit: mango
> |
```

Q. 3. Explain list functions and methods with examples.